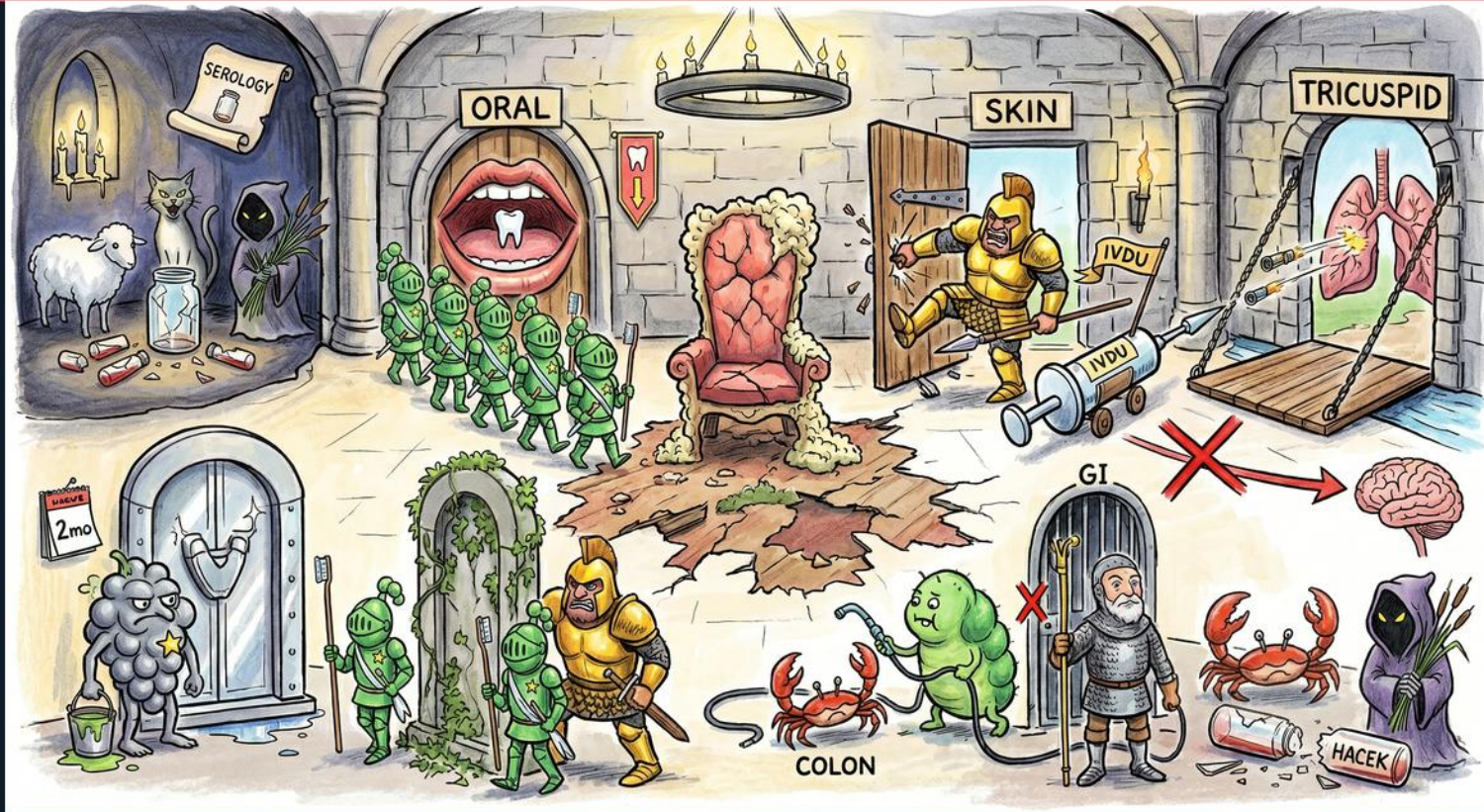


Infective Endocarditis — Organisms by Setting





Valve-throne vegetation

WHAT YOU SEE

The fleecy heart-valve throne with a cauliflower vegetation oozing onto the floor is the pathognomonic endocarditis lesion sitting on the leaflet.

WHAT IT ENCODES

The vegetation is a friable mass of platelets, fibrin, and microcolonies of organisms on a valve leaflet, sitting in an avascular niche where neutrophils and antibiotics penetrate poorly, so bacteria reach 10^9 – 10^{11} /gram. It forms on the atrial side of the mitral and ventricular side of the aortic valve (low-pressure side of the regurgitant jet) and drives both continuous bacteremia and embolization. Endothelial injury from turbulent jets first lays down a sterile thrombus that a transient bacteremia then colonizes.



ORAL door — viridans strep

WHAT YOU SEE

The open-mouth doorway with green knights marching out is the oral portal feeding viridans streptococci to the valve.

WHAT IT ENCODES

The oral cavity is the #1 portal (~40%). Viridans group strep (*S. sanguinis*, *S. mutans*, *S. mitis*) are alpha-hemolytic, optochin-RESISTANT oral flora that produce dextran to adhere to valves and cause SUBACUTE IE on a previously ABNORMAL valve after dental work or poor dentition. Penicillin-susceptible strains: penicillin G or ceftriaxone x4 weeks (2 weeks if gentamicin synergy added).



Green viridans army

WHAT YOU SEE

The marching column of little green knights is the indolent viridans-strep horde colonizing an already-damaged valve.

WHAT IT ENCODES

Viridans strep cause the classic indolent subacute course — low-grade fever, weight loss, night sweats — on rheumatic, myxomatous (MVP), bicuspid, or degenerative valves. They need a pre-roughened endothelium because, unlike *S. aureus*, they lack adhesins for intact endothelium. Optochin RESISTANCE separates viridans from *S. pneumoniae*.



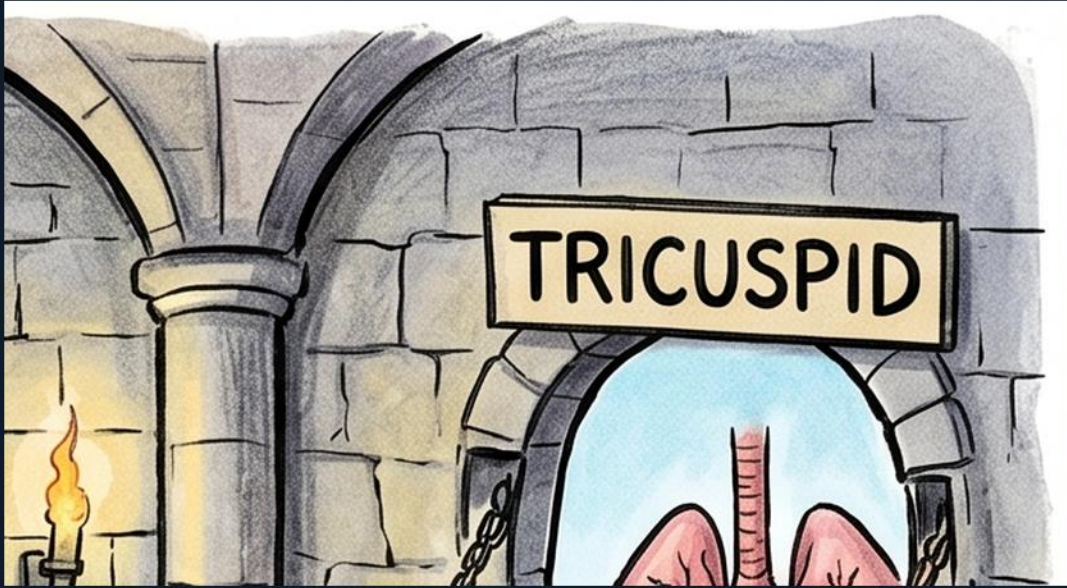
SKIN door — *S. aureus* / IVDU

WHAT YOU SEE

The gold-armored warrior bursting through the SKIN door clutching an IVDU syringe is *S. aureus* entering via skin and needles.

WHAT IT ENCODES

Skin/vascular access (~33%, second portal): IV drug use, central lines, hemodialysis catheters. *S. aureus* (GPC clusters, catalase+, coagulase+) is the #1 OVERALL cause and the leading healthcare-associated cause; its MSCRAMM adhesins (clumping factor, fibronectin-binding protein) let it invade NORMAL valves, producing ACUTE, rapidly destructive IE with abscess and embolism. Any *S. aureus* bacteremia mandates echo. MSSA → nafcillin/oxacillin/cefazolin; MRSA → vancomycin or daptomycin.



TRICUSPID lungs door

WHAT YOU SEE

The lung doorway labeled TRICUSPID with the syringe lodged at the valve shows right-sided IVDU disease seeding the lungs.

WHAT IT ENCODES

In people who inject drugs, *S. aureus* dominates and the TRICUSPID valve is hit first because venous blood reaches it first. Right-sided IE throws SEPTIC PULMONARY emboli — multiple peripheral cavitory nodules, pleuritic chest pain, cough — and is often murmurless. Right-sided *S. aureus* IE may respond to medical therapy alone unless vegetations are large or infection persists.



No systemic/brain emboli (red-X brain)

WHAT YOU SEE

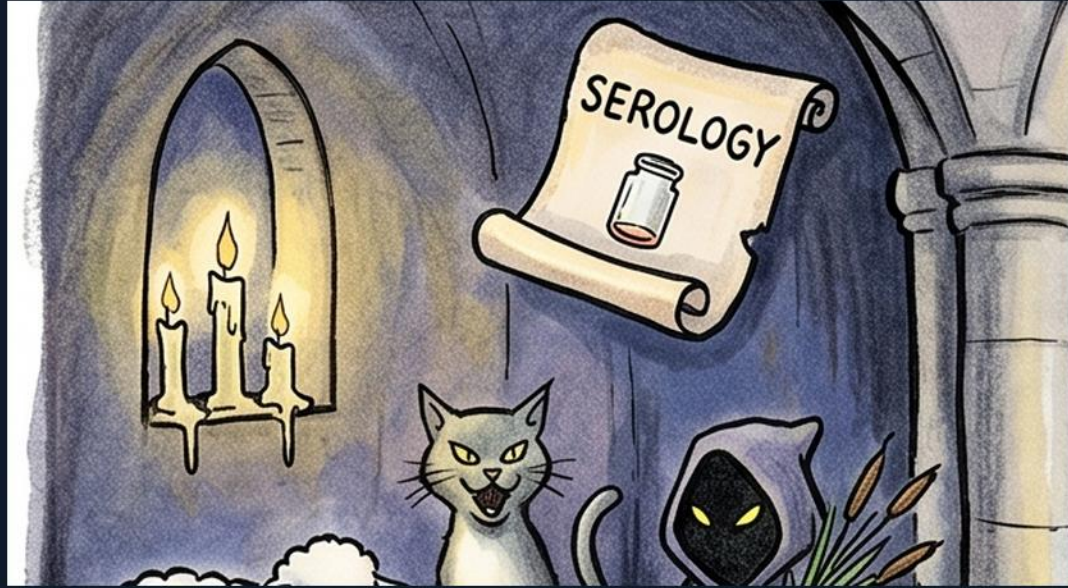
The brain crossed out by a red X warns that right-sided IVDU emboli do NOT travel to the brain or systemic circulation.

WHAT IT ENCODES

Right-sided (tricuspid) vegetations embolize to the LUNGS, not the systemic arteries, so brain/systemic emboli and a loud left-sided regurgitant murmur are NOT the IVDU picture. Chest CT with multiple cavitory nodules in an injector = septic pulmonary emboli.

THE BOARD TRAP

'IVDU endocarditis causes brain emboli and a loud regurgitant murmur' -> wrong; right-sided tricuspid disease seeds the LUNGS (septic pulmonary emboli), is frequently murmurless, and spares the systemic/cerebral circulation — that is exactly why the brain is X'd out.



SEROLOGY alcove (culture-negative)

WHAT YOU SEE

The shadowy candle-lit SEROLOGY alcove with sheep, cat, and grim reaper houses the blood-culture-NEGATIVE zoonotic organisms diagnosed by serology.

WHAT IT ENCODES

Culture-negative IE (~5–10%) comes from fastidious/intracellular/zoonotic bugs or prior antibiotics: *Coxiella* (Q fever — livestock), *Bartonella* (cats/homeless), *Brucella* (unpasteurized dairy), *Tropheryma whippelii*, fungi. Diagnosis is by SEROLOGY/PCR of valve tissue, not routine culture.

THE BOARD TRAP

'Order more blood cultures to grow the organism' -> wrong for these bugs; *Coxiella*, *Bartonella*, and *Brucella* are culture-NEGATIVE on routine media and are diagnosed by serology/PCR — repeating cultures just delays care.



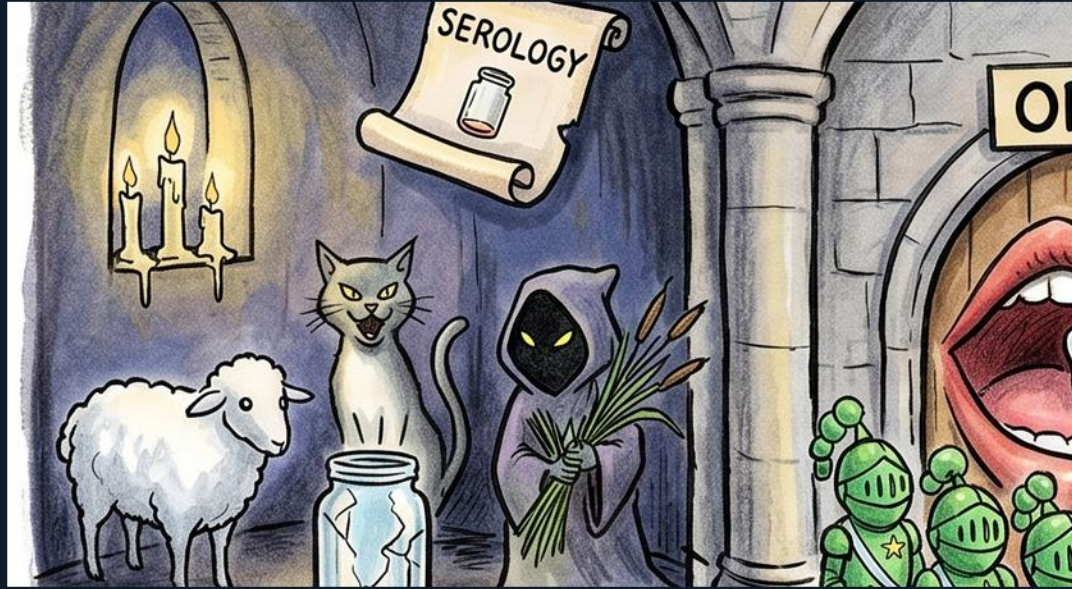
Coxiella — sheep / Q fever

WHAT YOU SEE

The sheep in the dark alcove anchors *Coxiella burnetii* (Q fever) from livestock exposure.

WHAT IT ENCODES

Coxiella burnetii (Q fever) is an intracellular, culture-negative cause of chronic IE linked to cattle/sheep/goats (parturient animals) and prosthetic valves. Phase I IgG $\geq 1:800$ is a STANDALONE major Duke criterion. Treat with doxycycline 100 mg BID + hydroxychloroquine (alkalinizes the phagolysosome so doxy becomes bactericidal) for 18–24 months with serial titers.



Bartonella — cat

WHAT YOU SEE

The cat beside the alcove anchors *Bartonella henselae* (cat scratch) and *B. quintana* (homeless / body louse).

WHAT IT ENCODES

Bartonella is a fastidious, culture-NEGATIVE gram-negative cause of IE: *B. henselae* from cats (scratch/flea), *B. quintana* from homelessness and the body louse. Diagnose by high IgG titer (major criterion) or valve PCR. Treat with doxycycline + gentamicin ± ceftriaxone; valve surgery is often required. Sort culture-negative IE by exposure: cats → *henselae*, homeless/lice → *quintana*, livestock → *Coxiella*.



Early PVE (<2 mo) — CoNS

WHAT YOU SEE

The grape-cluster bug with a gold star at the 2-month-calendar door is coagulase-negative staph seeding a fresh prosthesis.

WHAT IT ENCODES

Coagulase-NEGATIVE staph (chiefly *S. epidermidis*) is the #1 cause of EARLY prosthetic valve endocarditis (within ~2–12 months of surgery), seeded perioperatively and forming methicillin-resistant biofilm on the prosthetic material. Treat prosthetic staph IE with vancomycin + RIFAMPIN (biofilm penetration) + gentamicin for ≥ 6 weeks.

THE BOARD TRAP

'A single CoNS-positive culture is always a contaminant' -> wrong in a prosthetic-valve patient; CoNS is the leading EARLY PVE pathogen and the prime suspect, whereas in a native-valve patient one positive set is usually contamination.



Late PVE = community organisms

WHAT YOU SEE

The gold knight back among green knights at the ivy-covered door shows late PVE caused by the same community bugs as native valves.

WHAT IT ENCODES

LATE prosthetic valve endocarditis (>12 months postop) is caused by the SAME organisms as community native-valve IE — viridans strep, *S. aureus*, enterococci, HACEK — because the prosthesis is endothelialized and seeding is now hematogenous (dental, GU). Timing splits the differential: EARLY (<12 mo) = CoNS/*S. aureus* biofilm; LATE (>12 mo) = community flora.

THE BOARD TRAP

'All prosthetic-valve IE is coagulase-negative staph' -> wrong; only EARLY PVE is CoNS/*S. aureus* — LATE PVE mirrors community native-valve organisms (strep, enterococcus, HACEK).



S. gallolyticus → COLON

WHAT YOU SEE

The green germ holding a trident beside the crab labeled COLON is *S. gallolyticus* pointing to colorectal cancer.

WHAT IT ENCODES

Streptococcus gallolyticus subsp. *gallolyticus* (formerly *S. bovis* biotype I) is a GI-tract gram-positive coccus whose IE has a strong association with colorectal NEOPLASIA. Any *S. gallolyticus*/bovis bacteremia or IE MANDATES colonoscopy to exclude colorectal cancer. Mnemonic: 'bovis in the blood, cancer in the colon.'

THE BOARD TRAP

'Treat *S. bovis* IE with antibiotics and discharge' -> wrong; *S. gallolyticus* (bovis) IE obligates COLONOSCOPY to find an underlying colorectal cancer — skipping the cancer workup is the classic miss.



Enterococcus — GU source (red-X knight)

WHAT YOU SEE

The bearded gray knight with a trident and a red X over him is enterococcus from the GU tract — NOT a colon-cancer clue.

WHAT IT ENCODES

Enterococcus (*E. faecalis* >> *E. faecium*) is a GI/GU gram-positive coccus causing IE in OLDER MEN after genitourinary instrumentation (and women after obstetric procedures). It is intrinsically antibiotic-tolerant, so cure needs SYNERGISTIC combination — ampicillin + ceftriaxone (preferred, renal-sparing) or ampicillin + gentamicin, never monotherapy.

THE BOARD TRAP

'Young patient with enterococcal IE — look for colon cancer' -> wrong; the colon-cancer organism is *S. gallolyticus/bovis*. Enterococcus points to a GU/GI source in OLDER post-instrumentation patients, hence the red X.



HACEK — grim reaper / crabs

WHAT YOU SEE

The cloaked figure with crabs in the corner is the fastidious HACEK group of oral gram-negative coccobacilli.

WHAT IT ENCODES

HACEK = *Haemophilus*/*Aggregatibacter aphrophilus*, *Aggregatibacter actinomycetemcomitans*, *Cardiobacterium hominis*, *Eikenella corrodens*, *Kingella kingae* — fastidious gram-negative coccobacilli of oral/respiratory flora causing subacute IE, sometimes with large vegetations. First-line: ceftriaxone (they may make beta-lactamase). With modern automated culture systems they grow within the standard ~5-day incubation.

THE BOARD TRAP

'HACEK is culture-negative and needs weeks of special incubation' -> wrong; modern blood-culture systems recover HACEK within ~5 days, prolonged incubation is no longer required, and ceftriaxone is the treatment of choice.

Summary — every symbol

1. Valve-throne vegetation

The vegetation is a friable mass of platelets, fibrin, and microcolonies of organisms on a valve leaflet, sitting in an avascular niche where neutrophils and antibiotics penetrate poorly, so bacteria reach 10^9 – 10^{11} /gram

3. Green viridans army

Viridans strep cause the classic indolent subacute course — low-grade fever, weight loss, night sweats — on rheumatic, myxomatous (MVP), bicuspid, or degenerative valves

5. TRICUSPID lungs door

In people who inject drugs, S

7. SEROLOGY alcove (culture-negative)

Culture-negative IE (~5–10%) comes from fastidious/intracellular/zoonotic bugs or prior antibiotics: Coxiella (Q fever — livestock), Bartonella (cats/homeless), Brucella (unpasteurized dairy), Tropheryma whipplei, fungi

9. Bartonella — cat

Bartonella is a fastidious, culture-NEGATIVE gram-negative cause of IE: B

11. Late PVE = community organisms

LATE prosthetic valve endocarditis (>12 months postop) is caused by the SAME organisms as community native-valve IE — viridans strep, S

13. Enterococcus — GU source (red-X knight)

Enterococcus (E)

2. ORAL door — viridans strep

The oral cavity is the #1 portal (~40%)

4. SKIN door — S. aureus / IVDU

Skin/vascular access (~33%, second portal): IV drug use, central lines, hemodialysis catheters

6. No systemic/brain emboli (red-X brain)

Right-sided (tricuspid) vegetations embolize to the LUNGS, not the systemic arteries, so brain/systemic emboli and a loud left-sided regurgitant murmur are NOT the IVDU picture

8. Coxiella — sheep / Q fever

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10. Early PVE (<2 mo) — CoNS

Coagulase-NEGATIVE staph (chiefly S

12. S. gallolyticus → COLON

Streptococcus gallolyticus subsp

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